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ALLIGATOR CLIP

Abstract

An alligator clip has a clamping module and an insulating module. The clamping module has two clamping elements, an engaging portion, and an elastic element. The engaging structure is disposed between the two clamping elements to connect the two clamping elements with each other and has an engaging portion and a holding portion. The engaging portion is disposed on one of the two clamping elements and has a free end. The holding portion is disposed on the other one of the two clamping elements and engages with the free end of the engaging portion. The elastic element is disposed between the two clamping elements, abuts against the two clamping elements, and is disposed adjacent to the engaging structure. The insulating module is connected to the clamping module and has two outer covers respectively connected to the two clamping elements.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a clamp, and more particularly to an alligator clip that has a simplified structure, can be assembled easily, and can save cost.

2. Description of Related Art

[0002] With reference to FIGS. **12** and **13**, a conventional alligator clip **40** has a clamping module **41** and an insulating module **42**. The clamping module **41** has two clamping elements **43** and a torsion spring **44**. The two clamping elements **43** are connected to each other by a pivot shaft **45** and are moved toward or away from each other by using the pivot shaft **45** as a fulcrum. The torsion spring **44** is disposed between the two clamping elements **43** and abuts against the two clamping elements **43** to enable two ends of the two clamping elements **43** to remain in a state of contacting. Furthermore, the pivot shaft **45** passes through the torsion spring **44** and is connected to the two clamping elements **43** to form the clamping module **41** as an axial type clamping module. The insulating module **42** is connected to the clamping module **41** and has two outer covers **46** respectively connected to the two clamping elements **43** to provide an insulation effect. Then the conventional alligator clip **40** can be used to conduct electricity or signals.

[0003] However, with reference to FIGS. **12** and **14**, when assembling the conventional alligator clip **40**, since the two outer covers **46** are respectively mounted on the two clamping elements **43** to cover the two clamping elements **43**, if the two outer covers **46** are connected to the two clamping elements **43** first, the pivot shaft **45** cannot pass through the two clamping elements **43**. Therefore, with reference to FIG. **15**, when assembling the conventional alligator clip **40**, the pivot shaft **45** is passed through the two clamping elements **43** and the torsion spring **44** first and is riveted to hold the torsion spring **44** between the two clamping elements **43**. Furthermore, during assembly, in order to enable the two ends of the two clamping elements **43** to abut against each other, the torsion spring **44** is pressed against by the two clamping elements **43**. At this time, on one end, it is necessary to pass the pivot shaft **45** through the two clamping elements **43** and the torsion spring **44**, on the other end, it is necessary to press against the torsion spring **44** by the two clamping elements **43**, and this will increase time and inconvenience required for assembly.

[0004] Additionally, after assembling the clamping module **41**, the two outer covers **46** are respectively connected to the two clamping elements **43**. In order to avoid the two outer covers **46** separating from the two clamping elements **43** while the conventional alligator clip **40** is in use, the two outer covers **46** are securely connected to the two clamping elements **43** by hot melting. The hot melting is performed via inner sides of the two clamping elements **43** with the two outer covers **46**, and the two ends of the two clamping elements **43** need to be opened by a fixture, and this will increase time required for assembly and require expenses to buy the fixture to open the two clamping elements **43** and will increase cost for assembling the conventional alligator clip **40**. Therefore, the conventional alligator clip **40** needs to be improved.

[0005] The alligator clip in accordance with the present invention mitigates or obviates the aforementioned problems.

SUMMARY OF THE INVENTION

[0006] The main objective of the present invention is to provide an alligator clip that has a simplified structure, can be assembled easily, and can save cost.

[0007] The alligator clip in accordance with the present invention has a clamping module and an insulating module. The clamping module has two clamping elements, an engaging portion, and an elastic element. The engaging structure is disposed between the two clamping elements to connect the two clamping elements with each other and has an engaging portion and a holding portion. The engaging portion is disposed on one of the two clamping elements and has a free end. The holding portion is disposed on the other one of the two clamping elements and engages with the free end of the engaging portion. The elastic element is disposed between the two clamping elements, abuts against the two clamping elements, and is disposed adjacent to the engaging structure. The insulating module is connected to the clamping module and has two outer covers respectively connected to the two clamping elements.

[0008] Other objectives, advantages and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a perspective view of an alligator clip in accordance with the present invention;

[0010] FIG. **2** is a top exploded perspective view of the alligator clip in FIG. **1**;

[0011] FIG. **3** is a bottom exploded perspective view of the alligator clip in FIG. **1**;

[0012] FIG. **4** is an enlarged cross-sectional side view of the alligator clip in FIG. **1**;

[0013] FIG. **5** is another enlarged cross sectional side view of the alligator clip in FIG. **1**;

[0014] FIG. **6** is a perspective view of a clamping module of the alligator clip in FIG. **1**;

[0015] FIG. **7** is an enlarged perspective view in partial section of the clamping module of the alligator clip in FIG. **6**, omitting an elastic element;

[0016] FIG. **8** is an enlarged cross-sectional side view of the clamping module of the alligator clip in FIG. **6**, omitting the elastic element;

[0017] FIG. **9** is an operational cross-sectional side view of the alligator clip in FIG. **1**;

[0018] FIG. **10** is an operational perspective view of the clamping module of the alligator clip in FIG. **1**;

[0019] FIG. **11** is an operational perspective view in partial section of the clamping module of the alligator clip in FIG. **1**, omitting the elastic element;

[0020] FIG. **12** is an exploded perspective view of an alligator clip in accordance with the prior art;

[0021] FIG. **13** is a perspective view of a clamping module of the alligator clip in FIG. **12**;

[0022] FIG. **14** is a side view in partial section of the clamping module of the alligator clip in FIG. **13**; and

[0023] FIG. **15** is an enlarged side view in partial section of the clamping module of the alligator clip in FIG. **13**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0024] With reference to FIGS. **1** to **4**, an alligator clip in accordance with the present invention has a clamping module **10** and an insulating module **20**.

[0025] With reference to FIGS. **5** and **6**, the clamping module **10** has two clamping elements, an engaging structure **13**, and an elastic element **14**. The two clamping elements, respectively a first clamping element **11** and a second clamping element **12**, are made of a conductive material and are pivotally connected to each other. The engaging structure **13** is disposed between the first clamping element **11** and the second clamping element **12** to connect the second clamping element **12** with the first clamping element **11**. The engaging structure **13** has an engaging portion **15** and a holding portion **16**. The engaging portion **15** is disposed on one of the first clamping element **11** and the second clamping element **12** and has a free end, and the holding portion **16** is disposed on the other one of the first clamping element **11** and the second clamping element **12** and engages with the free end of the engaging portion **15**.

[0026] The elastic element **14** is disposed between the first clamping element **11** and the second clamping element **12**, abuts against the first clamping element **11** and the second clamping element

12, and is disposed adjacent to the engaging structure **13**. The first clamping element **11** and the second clamping element **12** are pushed by the elastic element **14** to enable two ends of the first clamping element **11** and the second clamping element **12** away from the elastic element **14** to abut each other.

[0027] Furthermore, each one of the first clamping element **11** and the second clamping element **12** has a connecting protrusion **17** disposed on an inner side of the clamping element, and two ends of the elastic element **14** are respectively mounted on the two connecting protrusions **17** of the first clamping element **11** and the second clamping element **12**. Then the elastic element **14** is stably disposed between the first clamping element **11** and the second clamping element **12**. Preferably, the elastic element **14** may be a compressed spring. Additionally, with reference to FIGS. **2** and **3**, each one of the first clamping element **11** and the second clamping element **12** has multiple assembling holes **111**, **121** formed through the clamping element at spaced intervals.

[0028] Preferably, with reference to FIGS. **7** and **8**, the engaging portion **15** of the engaging structure **13** is disposed on the first clamping element **11** and is formed by two engaging panels **18**. The two engaging panels **18** are disposed on the first clamping element **11** at a spaced interval by stamping and are elastically deformable. Each one of the two engaging panels **18** has an engaging end **181**, and the two engaging ends **181** of the two engaging panels **18** form the free end of the engaging portion **15**. The holding portion **16** of the engaging structure **13** is disposed on the second clamping element **12** and is formed by two engaging holes **19**. The two engaging holes **19** are disposed on the second clamping element **12** at a spaced interval by stamping and respectively engage with the two engaging panels **18**. Each one of the two engaging holes **19** has an inner surface abutted against the engaging end **181** of a corresponding one of the two engaging panels **18**. Then the first clamping element **11** and the second clamping element **12** are pivotally connected to each other by the two engaging panels **18** respectively engaging in the two engaging holes **19** without using a pivot shaft to form the clamping module **10** as a shaftless clamping module.

[0029] Furthermore, the locations of the engaging portion **15** and the holding portion **16** of the engaging structure **13** can be exchanged with each other, i.e., the engaging portion **15** is disposed on the second clamping element **12**, and the holding portion **16** is disposed on the first clamping element **11**, as long as the holding portion **16** engages with the engaging portion **15**.

[0030] With reference to FIGS. **2** to **5**, the insulating module **20** is connected to the clamping module **10** and has two outer covers. The two outer covers, respectively a first outer cover **21** and a second outer cover **22**, are made of an insulating material, and are respectively connected to the first clamping element **11** and the second clamping element **12**. Each one of the first outer cover **21** and the second outer cover **22** has multiple assembling tubes **211**, **221**. The multiple assembling tubes **211**, **221** of each one of the two outer covers are formed on and protruded from an inner side of the outer cover and are respectively extended through the multiple assembling holes **111**, **121** of a corresponding one of the two clamping elements. When the two outer covers are respectively connected to the two clamping elements, a hot melting method is used to connect the two outer covers securely with the two clamping elements. Preferably, the first outer cover **21** is securely connected to the first clamping element **11**, and the second outer cover **22** is securely connected to the second clamping element **12**.

[0031] With reference to FIGS. **9** and **10**, when the alligator clip of the present invention is in use, pressing the first outer cover **21** and the second outer cover **22** to enable the first clamping element **11** and the second clamping element **12** to move relative to each other by using the engaging structure **13** as a fulcrum to compress the elastic element **14**. Then the two ends of the first clamping element **11** and the second clamping element **12** away from the elastic element **14** are moved away from each other. At this time, moving the alligator clip of the present invention adjacent an electrical wire or a signal line and releasing to press the first outer cover **21** and the second outer cover **22**, and the compressed elastic element **14** will push the first clamping element **11** and the second clamping element **12** to move to enable the two ends of the first clamping

element **11** and the second clamping element **12** to move toward each other to clamp the electrical wire or the signal line. After using the alligator clip of the present invention, a user only needs to press the first outer cover **21** and the second outer cover **22** again to separate the first clamping element **11** and the second clamping element **12** from the electrical wire or the signal line.
[0032] With further reference to FIGS. **2** and **3**, to assemble the alligator clip of the present invention, since the first clamping element **11** and the second clamping element **12** can be connected with each other without using a pivot shaft, the first outer cover **21** and the second outer cover **22** can respectively and securely connect to the first clamping element **11** and the second clamping element **12** by hot melting, firstly. After the first outer cover **21** and the second outer cover **22** are securely connected to the first clamping element **11** and the second clamping element **12**, the elastic element **14** is disposed between the two connecting protrusions **17** of the first clamping element **11** and the second clamping element **12**. At this time, with reference to FIG. **5**, moving the second clamping element **12** toward the first clamping element **11** enables the two engaging ends **181** of the engaging panels **18** of the engaging portion **15** to respectively engage in the two engaging holes **19** of the holding portion **16** to complete the assembly of the alligator clip of the present invention, and this is convenient in assembly, can save time of assembly, and there is no need to buy a fixture to open the first clamping element **11** and the second clamping element **12**, reducing cost of using the alligator clip. Therefore, the present invention provides an alligator clip that has a simplified structure, can be assembled easily, and can save cost.
[0033] Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the utility model, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An alligator clip comprising: a clamping module having two clamping elements; an engaging structure disposed between the two clamping elements to connect the two clamping elements with each other and having an engaging portion disposed on one of the two clamping elements and having a free end; and a holding portion disposed on the other one of the two clamping elements and engaging with the free end of the engaging portion; and an elastic element disposed between the two clamping elements, abutted against the two clamping elements, and disposed adjacent to the engaging structure; and an insulating module connected to the clamping module and having two outer covers respectively connected to the two clamping elements.
2. The alligator clip as claimed in claim 1, wherein the engaging portion of the engaging structure is formed by two engaging panels; the two engaging panels are disposed on one of the two clamping elements at a spaced interval and are elastically deformable; each one of the two engaging panels has an engaging end, and the two engaging ends of the two engaging panels form the free end of the engaging portion; the holding portion of the engaging structure is formed by two engaging holes; the two engaging holes are disposed on the other one of the two clamping elements at a spaced interval and respectively engage with the two engaging panels; and each one of the two engaging holes has an inner surface abutted against the engaging end of a corresponding one of the two engaging panels.
3. The alligator clip as claimed in claim 2, wherein each one of the two clamping elements has multiple assembling holes formed through the clamping element at spaced intervals; and each one of the two outer covers has multiple assembling tubes formed on and protruded from an inner side of the outer cover and respectively extended through the multiple assembling holes of a corresponding one of the two clamping elements.
4. The alligator clip as claimed in claim 2, wherein the two clamping elements are respectively a

first clamping element and a second clamping element; the two outer covers are respectively a first outer cover and a second outer cover; and the first outer cover is connected to the first clamping element and the second outer cover is connected to the second clamping element.

5. The alligator clip as claimed in claim 3, wherein the two clamping elements are respectively a first clamping element and a second clamping element; the two outer covers are respectively a first outer cover and a second outer cover; and the first outer cover is connected to the first clamping element and the second outer cover is connected to the second clamping element.

6. The alligator clip as claimed in claim 4, wherein each one of the two clamping elements has a connecting protrusion disposed on an inner side of the clamping element; and the elastic element has two ends respectively mounted on the connecting protrusions of the two clamping elements.

7. The alligator clip as claimed in claim 5, wherein each one of the two clamping elements has a connecting protrusion disposed on an inner side of the clamping element; and the elastic element has two ends respectively mounted on the connecting protrusions of the two clamping elements.

8. The alligator clip as claimed in claim 1, wherein the elastic element is a compressed spring.

9. The alligator clip as claimed in claim 2, wherein the elastic element is a compressed spring.

10. The alligator clip as claimed in claim 3, wherein the elastic element is a compressed spring.

11. The alligator clip as claimed in claim 1, wherein the two clamping elements are respectively a first clamping element and a second clamping element; the two outer covers are respectively a first outer cover and a second outer cover; and the first outer cover is connected to the first clamping element and the second outer cover is connected to the second clamping element.

12. The alligator clip as claimed in claim 11, wherein the engaging portion of the engaging structure is formed by two engaging panels; the two engaging panels are disposed on one of the two clamping elements at a spaced interval and are elastically deformable; each one of the two engaging panels has an engaging end, and the two engaging ends of the two engaging panels form the free end of the engaging portion; the holding portion of the engaging structure is formed by two engaging holes; the two engaging holes are disposed on the other one of the two clamping elements at a spaced interval and respectively engage with the two engaging panels; and each one of the two engaging holes has an inner surface abutted against the engaging end of a corresponding one of the two engaging panels.

13. The alligator clip as claimed in claim 11, wherein each one of the two clamping elements has multiple assembling holes formed through the clamping element at spaced intervals; and each one of the two outer covers has multiple assembling tubes formed on and protruded from an inner side of the outer cover and respectively extended through the multiple assembling holes of a corresponding one of the two clamping elements.

14. The alligator clip as claimed in claim 12, wherein each one of the two clamping elements has multiple assembling holes formed through the clamping element at spaced intervals; and each one of the two outer covers has multiple assembling tubes formed on and protruded from an inner side of the outer cover and respectively extended through the multiple assembling holes of a corresponding one of the two clamping elements.

15. The alligator clip as claimed in claim 13, wherein the elastic element is a compressed spring.

16. The alligator clip as claimed in claim 14, wherein the elastic element is a compressed spring.